

# Data Visualization

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# Definition

Data visualization is the graphical representation of information and data.

-> An important step after collection, observation and analysis of data

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# History

- The concept of using pictures to understand data has been around for centuries, from maps and graphs in the 17th century to the invention of the pie chart in the early 1800s.
  - One of the most cited examples of statistical graphics occurred when Charles Minard mapped Napoleon's invasion of Russia. The map depicted the size of the army as well as the path of Napoleon's retreat from Moscow – and tied that information to temperature and time scales for a more in-depth understanding of the event.
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# Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

Dressée par M. Minard, Inspecteur Général des Ponts et Chaussées en retraite. Paris, le 20 Novembre 1869.

Les nombres d'hommes présents sont représentés par les largeurs des zones colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en travers des zones. Le rouge désigne les hommes qui entrent en Russie; le noir ceux qui en sortent. Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M. M. Chiers, de Legur, de Fezensac, de Chambray et le journal inédit de Jacob, pharmacien de l'Armée depuis le 28 Octobre.

Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Jérôme et du Maréchal Davout, qui avaient été détachés sur Minsk et Mohilew et qui rejoignent Orscha et Witebsk, avaient toujours marché avec l'armée.

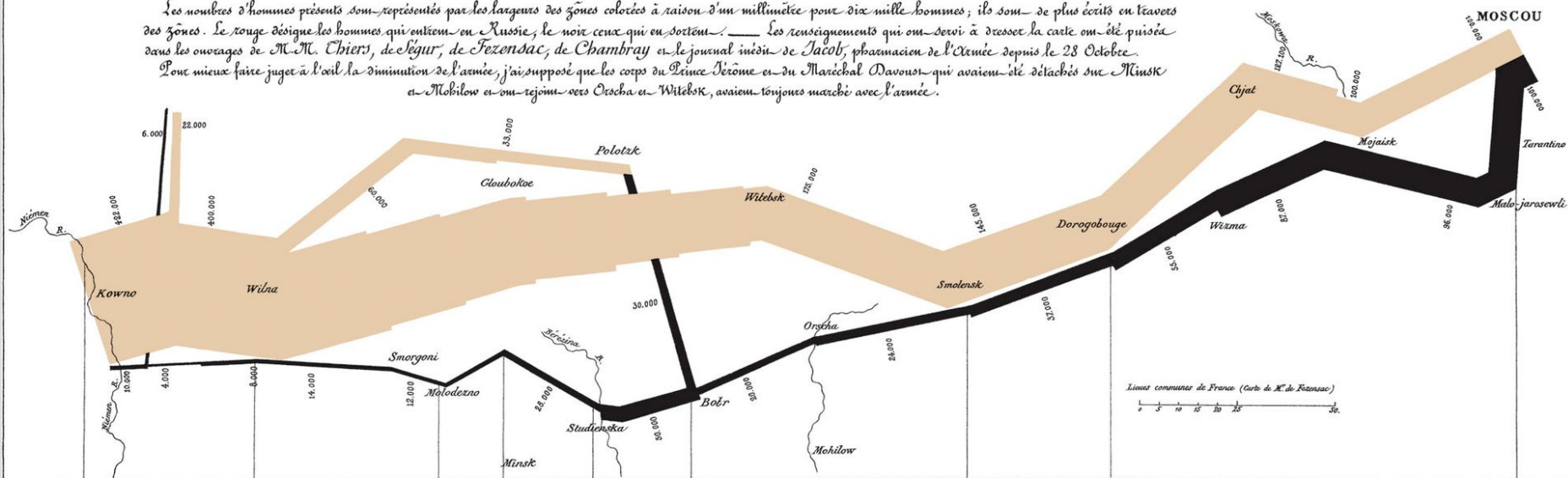
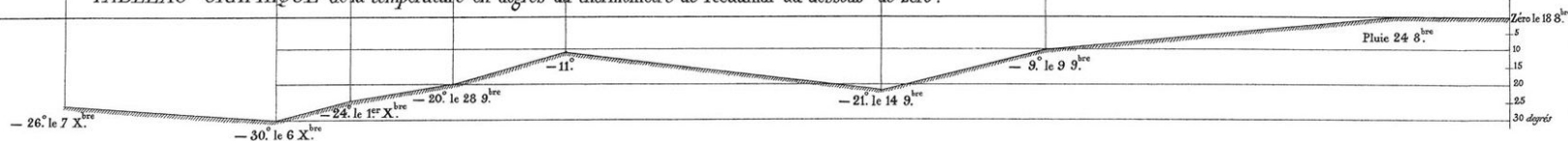
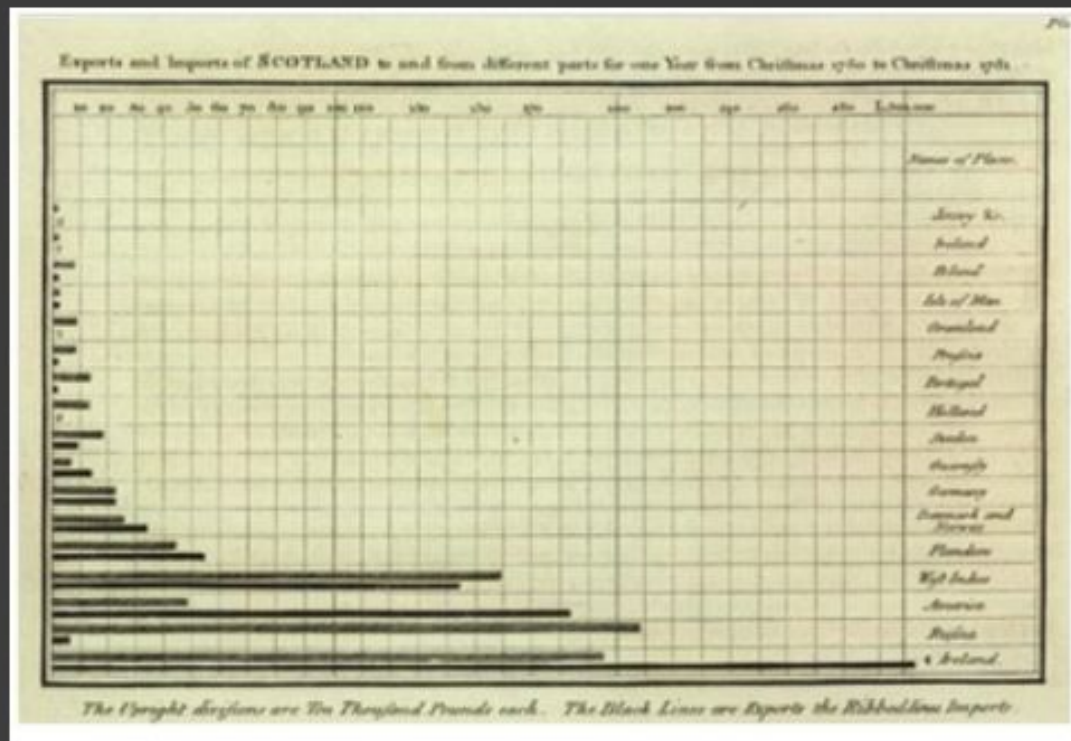


TABLEAU GRAPHIQUE de la température en degrés du thermomètre de Réaumur au dessous de zéro.

Les Cosaques passent au galop le Niémen gelé.





1781

William Playfair creates the first bar chart, representing Scotland's imports and exports from 17 countries. He strongly believed charts communicated better than tables of data, and is also credited with inventing the line, area, and pie charts.

**Why is Data Visualization  
important?**

# 2.5 quintillion bytes

According to the World Economic Forum, the world produces this much amount of data every day, and 90% of all data has been created in the last two years.

**“The better you can convey your points visually, the better you can leverage that information.”**

- Data Visualization makes it easy to see and understand large data
- To detect patterns, trends and outliers
- Improved insight
- For meaningful storytelling
- Data Visualization plays an important role in big data and advanced analytics projects
- To identify areas that need attention or improvement
- To Clarify which factors influence behavior of data or consumers
- And finally, to Predict



**Some interesting applications**

**To depict or represent**



#### COMMAND/SERVICE MODULES

- 1 GUMDROP
- 2 CHARLIE BROWN
- 3 COLUMBIA
- 4 YANKEE CLIPPER
- 5 ODYSSEY
- 6 KITTY HAWK
- 7 ENDAVOUR
- 8 CASPER
- 9 AMERICA

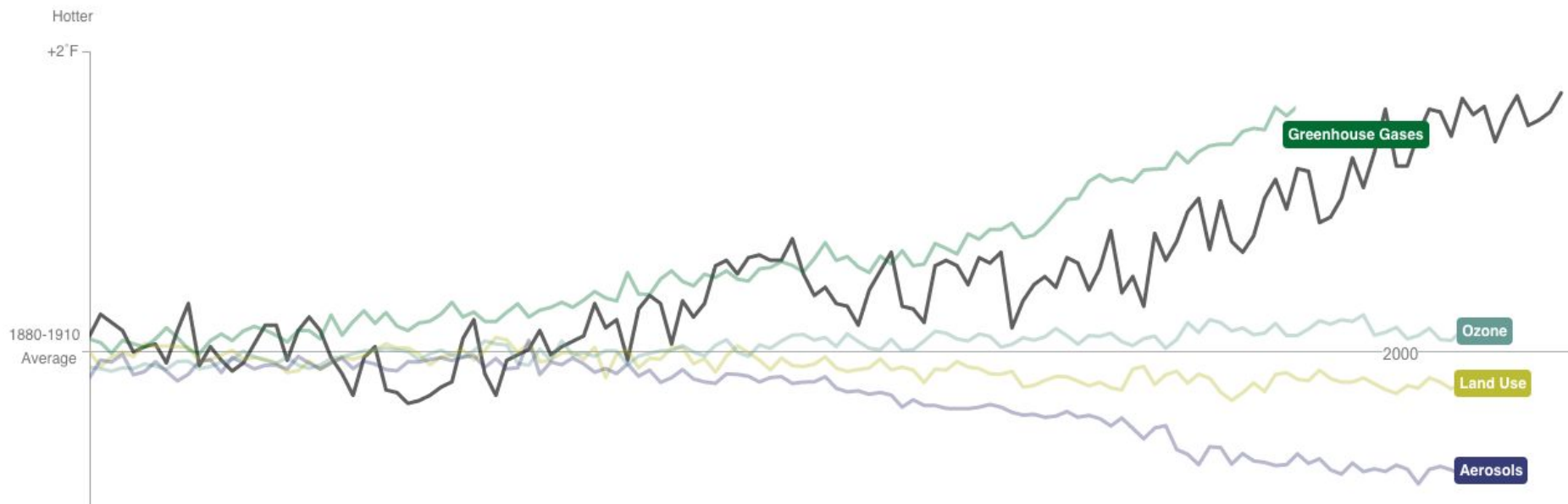


#### LUNAR MODULES

- 10 SPIDER
- 11 ENDORF
- 12 EAGLE
- 13 INTREPID
- 14 AQUARIUS
- 15 ANTARES
- 16 FALCON
- 17 ORION
- 18 CHALLENGER

## See for Yourself

Greenhouse gases warm the atmosphere. Aerosols cool it a little bit. Ozone and land-use changes add and subtract a little. Together they match the observed temperature, particularly since 1950.



# THE MANY MOONS OF JUPITER

**IN** 1610, Galileo Galilei gazed up through his telescope in the direction of Jupiter. In that moment he likely became the first person to see a moon other than our own, as light that had left the vicinity of the gas giant around half an hour earlier crashed into his pupils and revealed four dotted silhouettes. These Galilean moons, one of which is even larger than the planet Mercury, became the opening entry into a collection that is still increasing today. In fact in 2018, 407 years after the Italian polymath made his discovery, scientists confirmed the existence of

12 more moons locked in slow rotation with the largest planet in our solar system. These newly found satellites form part of a diverse family, many of which share little commonality other than their gravitational anchor. Their orbital shapes range from near perfect circles to highly eccentric and inclined. Their scales vary hugely, from the size of planets to just a kilometer across. Some may have been asteroids captured by Jupiter's powerful gravitational pull, while others were likely a by-product of the very formation of the planet itself.

This data visualization displays every currently known moon of Jupiter, each featuring the year of discovery, discoverer and a representation of scale. Additionally, on the right are some additional insights about the moons. Finally, while all information is correct as of 2018, scientists are finding new wonders in our solar system every day; so who knows how many new Jovian moons are out there right now, held in endless revolutions, just waiting for eyes to meet them for the first time?

**79**  
Moons discovered as of 2018

The number of moons that are prograde

The number of moons that are retrograde

The Voyager probes, launched in 1977, discovered three of Jupiter's large inner moons

**99.997%**  
The approx. % of the total mass in orbit around Jupiter that comes from the four Galilean moons

One of the newly discovered moons has an odd prograde orbit which sees its path cross several other retrograde moons. This means a collision is very likely, although scientists predict it could take another billion years to actually happen

As of 2018, five of the moons are considered lost

Callisto, Jupiter's largest moon, is actually 4% larger than the planet Mercury, making it the 16th largest object in our solar system

It's thought that several of the larger moons could feature subsurface oceans, leading to some exciting possibilities about the existence of life there

**54**  
The most moons have been found by a team led by Scott Sutherland

A German astronomer called Simon Marius independently discovered the four Galilean moons at the same time as Galileo. While he didn't receive the title of discoverer, he is responsible for their names, which are still used today

## Reading the data visualization

**Representing each moon**

Regular satellites

- Amalthea group
- Galilean moons

Irregular satellites

- Themis group
- Pasiphae group
- Amalthea group
- Callisto group
- Not part of a known family

Orbital direction

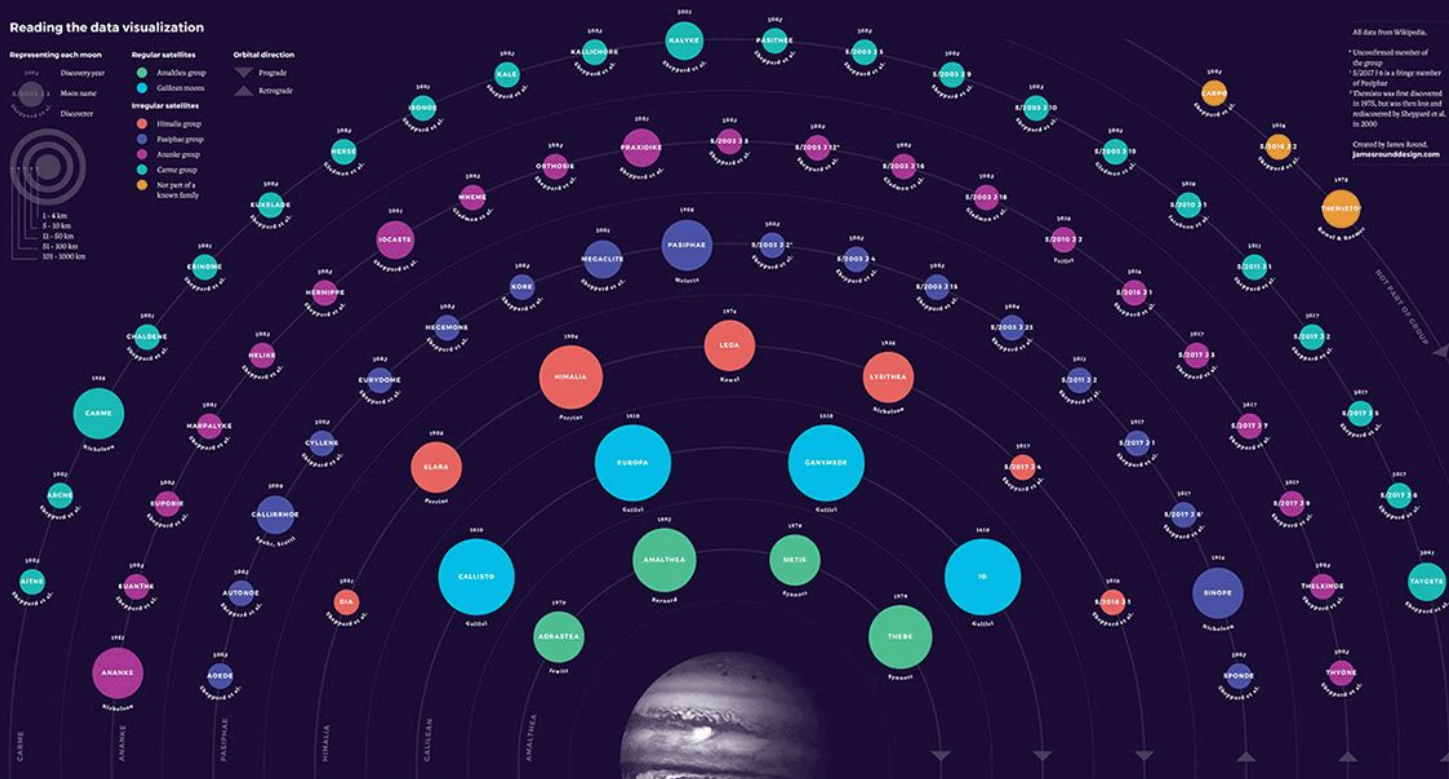
- Prograde
- Retrograde

Discover year

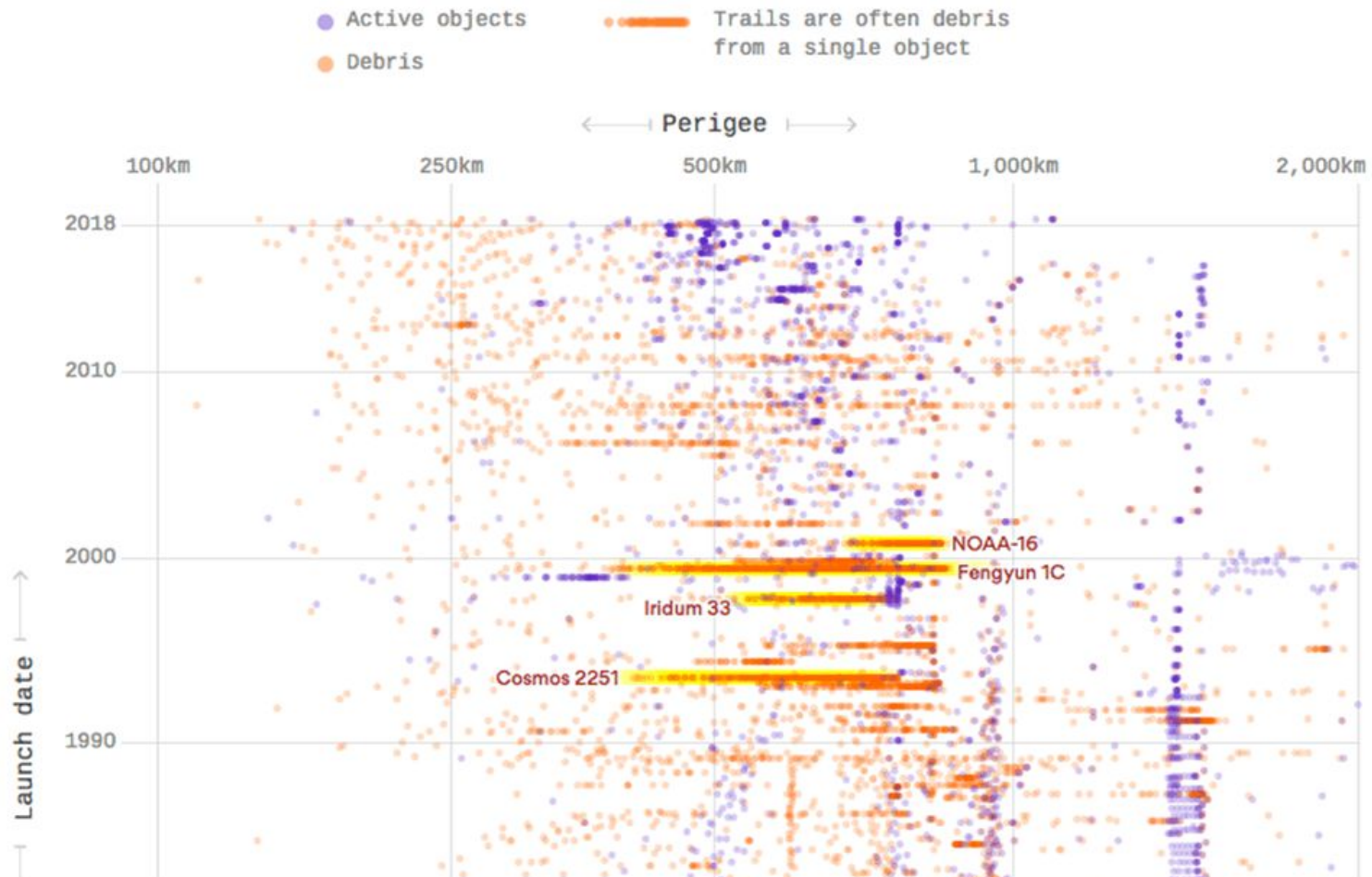
Discoverer

Scale

- 1 - 1 km
- 1 - 10 km
- 11 - 50 km
- 51 - 100 km
- 101 - 200 km
- 201 - 300 km



## Objects in low-earth orbit by launch date and perigee





SORT BY

**SPORT**

VIEW ALL

AMERICAN FOOTBALL

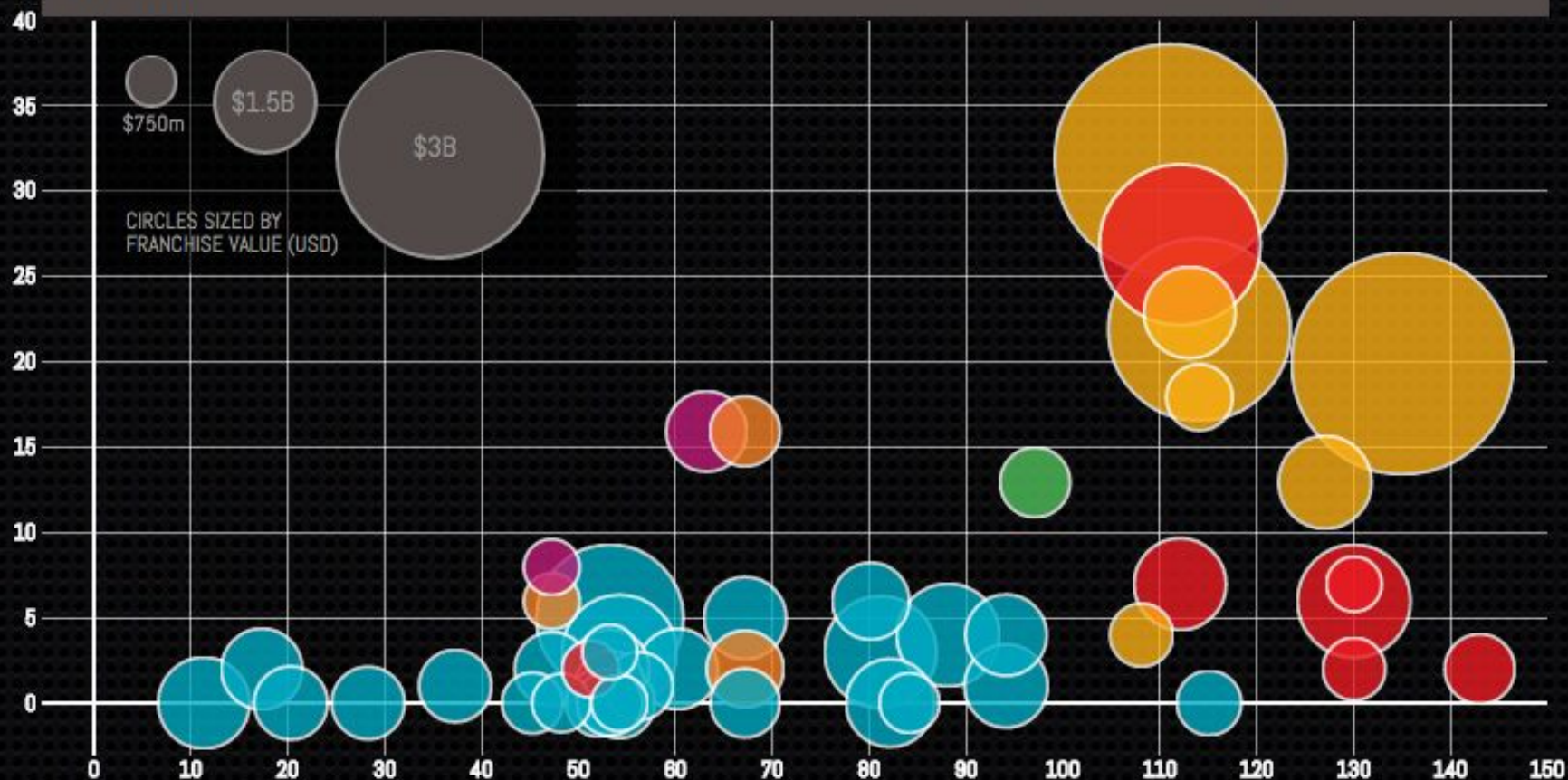
FORMULA ONE

BASEBALL

BASKETBALL

SOCCER/FOOTBALL

HOCKEY





This is a rendering of how the moon would look if it was filled with Earth's colors. A spectacular effort using the moon's topography which was converted to colormap using *matplotlib* and *cpt city*. The 3-D model was created with *blender* and *python*.

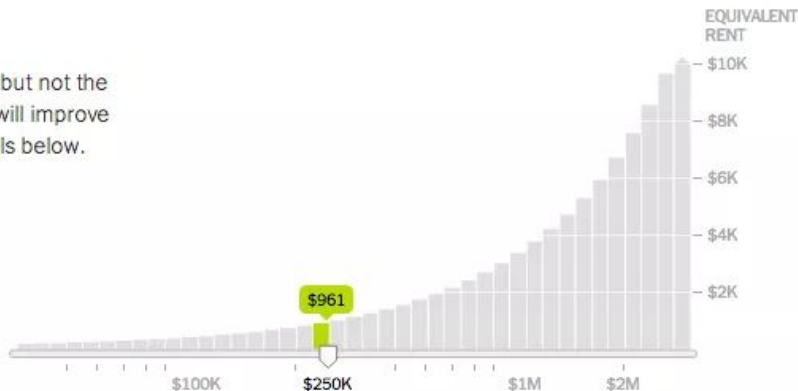


**To determine trends and make  
informed decisions**

## Home Price

A very important factor, but not the only one. Our estimate will improve as you enter more details below.

\$250,000



If you can rent a similar home for less than ...

**\$961** PER MONTH

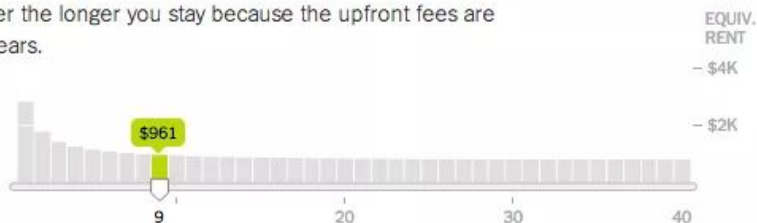
... then renting is better.

| Costs after 9 years | Rent      | Buy        |
|---------------------|-----------|------------|
| Initial costs       | \$961     | \$60,000   |
| Recurring costs     | \$116,288 | \$169,889  |
| Opportunity costs   | \$16,742  | \$45,506   |
| Net proceeds        | -\$961    | -\$142,366 |
| Total               | \$133,029 | \$133,029  |

## How Long Do You Plan to Stay?

Buying tends to be better the longer you stay because the upfront fees are spread out over many years.

9 years



## What Are Your Mortgage Details?

In addition to the interest rate and down payment, the calculator takes into account the mortgage-interest tax deduction.

EQUIV.  
RENT  
-\$4K

**How to Read the Charts** Charts that are relatively flat indicate factors that are not particularly important to the outcome. Conversely, the factors that have steep slopes have a large impact.

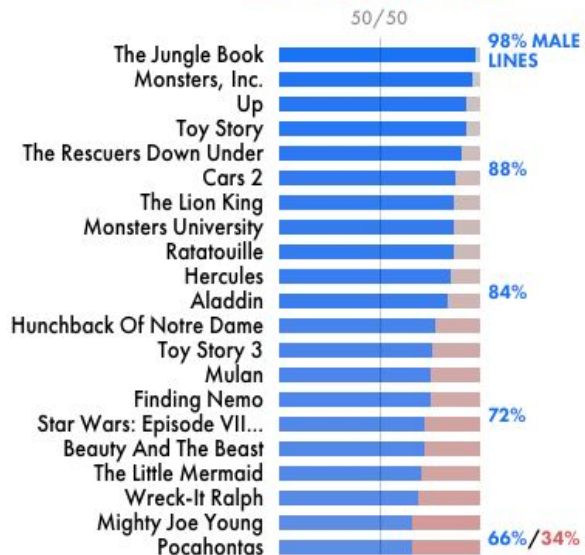


## Screenplay Dialogue, Broken-down by Gender

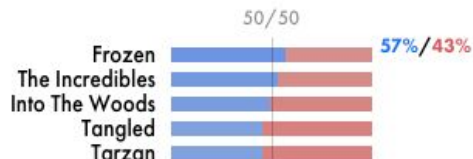
2,005 Screenplays: Dialogue  
Broken-down by Gender

Only High-Grossing Films: Ranked in  
the Top 2,500 by US Box Office \*

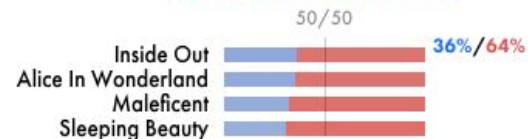
### Men have 60%+ Lines



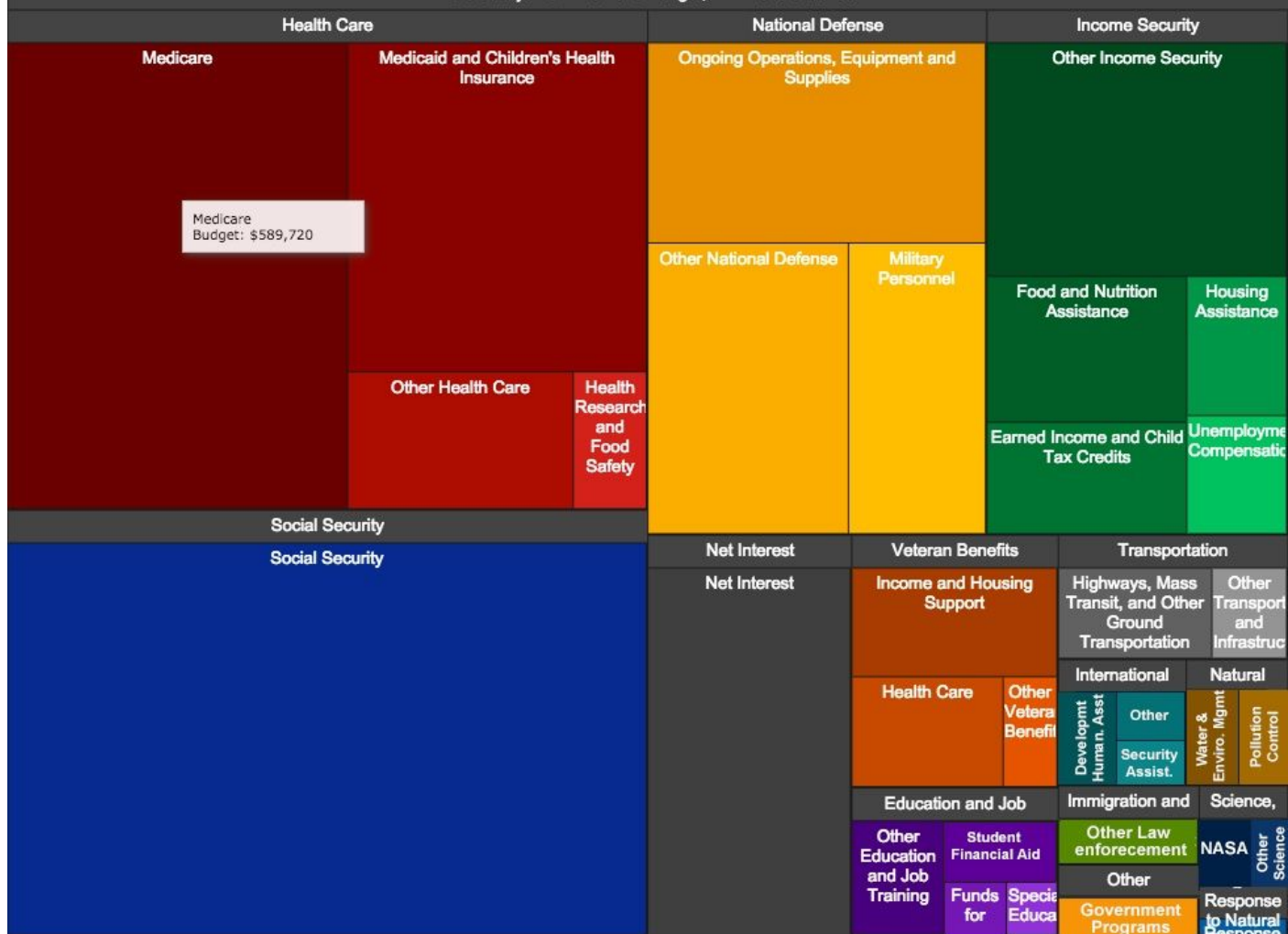
### Gender Balance, +/- 10%



### Women have 60%+ Lines



2016 outlays from the 2016 Budget, in millions of dollars



# NEXT AMERICA

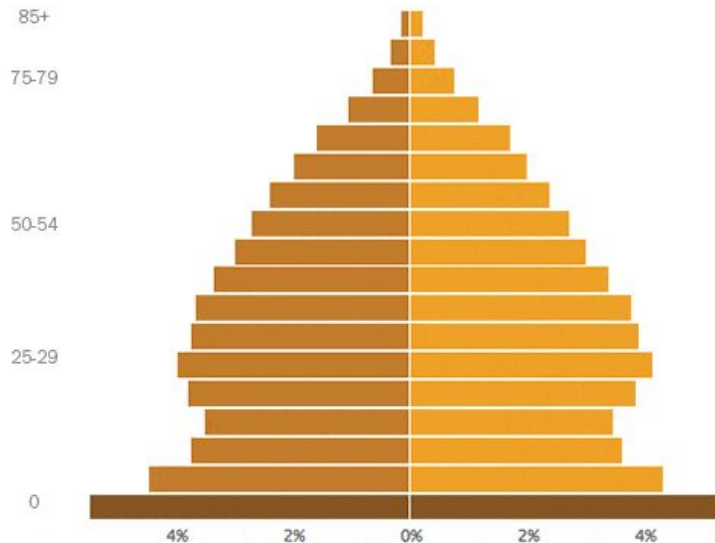
Percent of U.S. Population by Age Group, 1950-2060

Baby Boomers

MALE

1950

FEMALE



PEW RESEARCH CENTER

# NEXT AMERICA

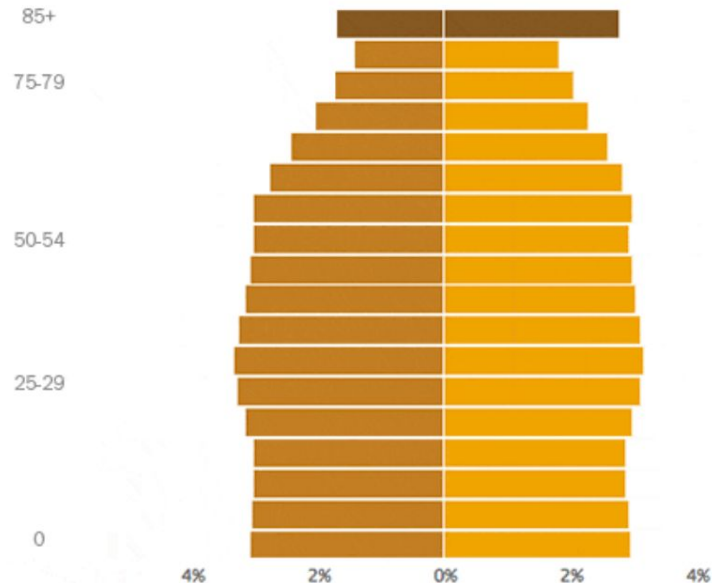
Percent of U.S. Population by Age Group, 1950-2060

Baby Boomers

MALE

2050

FEMALE



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# Some Challenges

- Data Availability
- Deciding which visual is the best

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